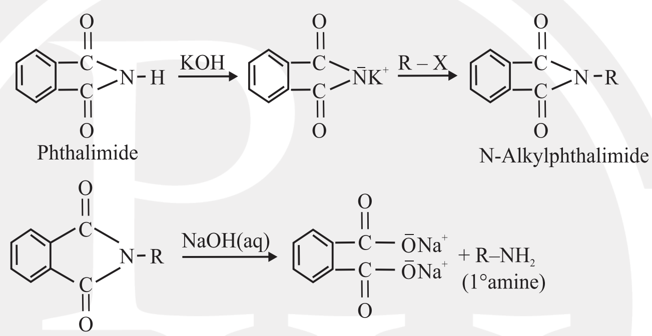
CHAPTER 7

Amines

Sr. No.	Concept	Description	Other information
1.	**Name reactions** (a) Gabriel phthalimide synthesis		Reagent = KOH, NaOH(aq.)
	(b) Hoffmann bromamide degradation reaction	$R-\overset{O}{\overset{\Vert}{C}}-NH_2 + Br_2 + 4NaOH \longrightarrow R-NH_2 + NH_2CO_3 + 2NaBr + 2H_2O$	Reagent = Br_2, NaOH(aq.)
	(c) Acylation	$C_6H_5-\underset{\mid \atop H}{\ddot{N}}-H + CH_3-\underset{\Vert \atop O}{C}-O-\underset{\Vert \atop O}{C}-CH_3 \longrightarrow C_6H_5-\underset{\mid \atop H}{\ddot{N}}-\underset{\Vert \atop O}{C}-CH_3 + CH_3COOH$ Benzenamine, Ethanoic anhydride N-Phenylethanamide or Acetanilide	
	(d) Carbylamine reaction	$R-NH_2 + CHCl_3 + 3KOH \xrightarrow{Heat} R-NC + 3KCl + 3H_2O$	Reagent = $CHCl_3$ & KOH It is used as a test for primary amines
	(e) Diazotisation	$C_6H_5NH_2 + NaNO_2 + 2HCl \xrightarrow{273 - 278K} C_6H_5\overset{+}{N_2}\overset{-}{Cl} + NaCl + 2H_2O$	Reagent = $NaNO_2$ & HCl

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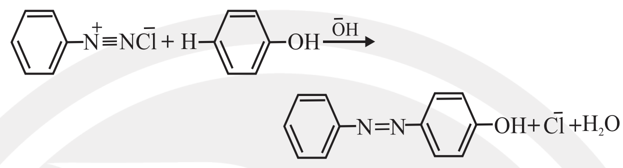
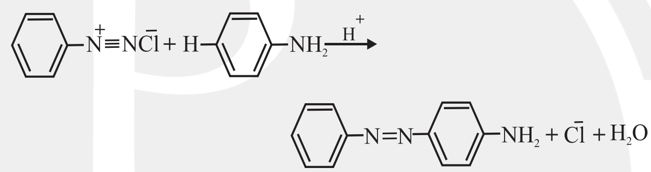
	(f) Sandmeyer reaction	$Ar\overset{+}{N}_2\overset{-}{X} \xrightarrow{Cu_2Cl_2/HCl} ArCl + N_2$ $Ar\overset{+}{N}_2\overset{-}{X} \xrightarrow{Cu_2Br_2/HBr} ArBr + N_2$ $Ar\overset{+}{N}_2\overset{-}{X} \xrightarrow{CuCN/KCN} ArCN + N_2$	Reagent = Cu_2Cl_2/HCl, Cu_2Br_2/HBr, $CuCN/KCN$
	(g) Gatterman reaction	$Ar\overset{+}{N}_2\overset{-}{X} \xrightarrow{Cu/HCl} ArCl + N_2 + CuX$ $Ar\overset{+}{N}_2\overset{-}{X} \xrightarrow{Cu/HBr} ArBr + N_2 + CuX$	Reagent = Cu/HCl, Cu/HBr
	(h) Coupling reactions	 *p*-Hydroxyazobenzene (orange dye)  *p*-Aminoazobenzene (yellow dye)	Reagent = H^+, OH^-
2.	**Physical properties of Amines** (a) Physical state	(i) Lower aliphatic amines $\longrightarrow$ gases with fishy smell (ii) Primary amine with 3 or more carbon atoms $\longrightarrow$ liquids with fishy odour (iii) Lower aromatic amines $\longrightarrow$ liquids with characteristic unpleasant odour (iv) Higher aromatic amines $\longrightarrow$ solids & odourless	
	(b) Solubility	$C_6H_5NH_2 < (C_2H_5)_2NH_2 < C_2H_5NH_2$	Solubility $\propto \frac{1}{\text{Molar mass}}$ Solubility $\propto$ H-bonding
	(c) Boiling point	1° amines > 2° amines > 3° amines	B.P. $\propto$ Mwt. B.P. $\propto$ H-bonding
	(d) Basicity order of Amines in aqueous medium	$(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5NH_2 > NH_3$ $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$	

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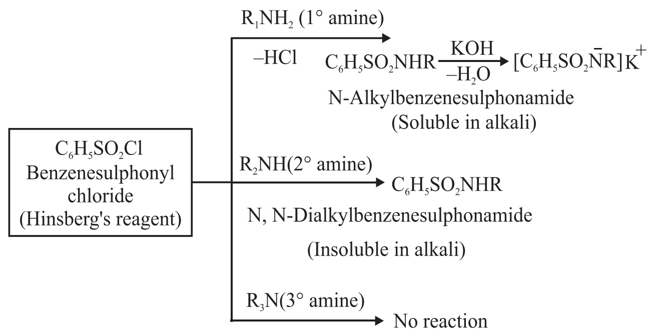
3.	**Distinction between 1°, 2° & 3° Amines** (a) Hinsberg's test		Reagent = HCl, KOH
	(b) Nitrous test (i) Primary aliphatic amines react with nitrous acid	$R - NH_2 + HNO_2 \xrightarrow{NaNO_2 + HCl} [R - \overset{+}{N}_2\overset{-}{Cl} \xrightarrow{H_2O} ROH + N_2 + HCl]$	Reagent = $NaNO_2$ + HCl
	(ii) Aromatic amines react with nitrous acid	$\underset{\text{Aniline}}{C_6H_5-NH_2} \xrightarrow[273-278\,K]{NaNO_2 + 2HCl} \underset{\text{Benzenediazonium chloride}}{C_6H_5 - \overset{+}{N}_2\overset{-}{Cl}} + NaCl + 2H_2O$	Reagent = $NaNO_2$ + HCl